

Grasp Planning for Robotic Manipulation in 6-DoF

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Abstract

Robotic grasping is a fundamental capability in manufacturing, healthcare, and domestic environments, enabling robots to manipulate objects efficiently and autonomously. This paper focuses on task-oriented grasping (TOG) in six degrees of freedom, which improves accuracy and translatability to real-world applications. We aim to contribute to the ongoing development of TOG by identifying challenges, evaluation metrics, describing two TOG approaches, and suggesting future research directions towards fully autonomous robotic task execution.

Keywords: Robotic Grasping, 6-DoF, Task-Oriented Grasping

1. Introduction

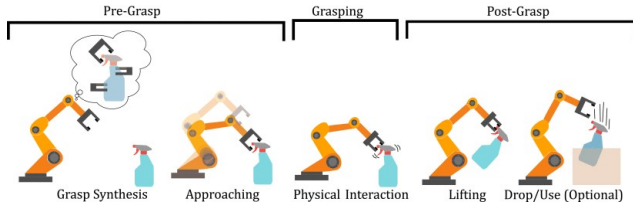


Figure 1. Typical stages for grasping an object. Image taken from [20].

Grasping is the most common feature of robots in manufacturing and daily life. It is crucial for most object manipulation tasks. Robots with grasping ability can improve the current labor shortage situation. They have a wide range of applications, from agriculture and medical care to warehousing and domestic work. Robots with grasping ability can perform tasks more efficiently and accurately than humans [9, 20, 23, 24].

The typical stages for grasping an object are displayed in Figure 1:

1. **Pre-Grasp:** First, Grasp Synthesis is applied to find a high-quality grasp. Grasp synthesis is the problem of creating a grasp. Usually, multiple grasp hypotheses are created and evaluate on certain criteria. Possible

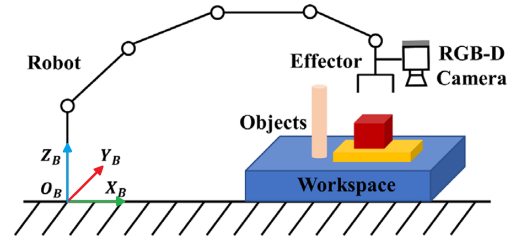


Figure 2. The robotic grasping setup. The robot arm consists of multiple kinematic elements and the end effector is equipped with a RGB-D camera. Image taken from [9].

criteria can be grasp stability, reliability, gripper geometry or kinematics [20, 22]. Second, Grasp Planning is applied. Grasp Planning is considered the trajectory planning of the gripper under certain conditions such as avoiding collision between the gripper and other objects [2].

2. **Grasping:** Grasping is defined as applying forces and torques at a set of contact to pick up an object [20]. In this stage the target object is grasped with the grasp found in the Grasp Synthesis.
3. **Post-Grasp:** In the Post-Grasp phase the object is usually lifted and optionally used for a specific task. [20].

Figure 2 shows an environment setup for robotic grasping. The robotic arm consists of several kinematic elements, the last of them is called the end effector. Often a camera is attached to the end effector. Depending on the algorithm, the camera captures RGB or RGB-D images. In both cases, object detection is applied to the input image to extract the target object. In the next step, the target object is passed to the previously mentioned grasping pipeline [9, 20]. Previous research has mostly used four degrees of freedom (4-DoF), while today six degrees of freedom (6-DoF) are also used for the end-effector space. 4-DoF considers the gripper poses in three dimensions plus the gripper orientation. It is often referred to as "top-down grasping" because the hand orientation vector is usually aligned with the direction

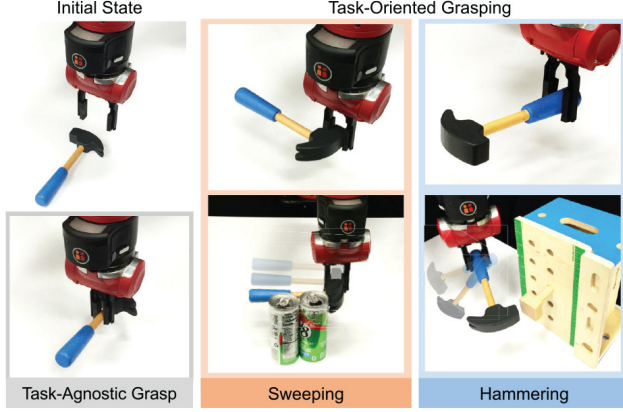


Figure 3. The same object might be grasped differently by a robot. A task-agnostic grasp might not be suitable for specific tasks such as sweeping or hammering. Image taken from [10].

of gravity. On the other hand, 6-DoF is considered to be the 3D position and orientation of the gripper. In comparison, 6-DoF offers more flexibility, but comes with increased complexity [2, 20].

Task-oriented approaches aim to understand how an object is used in a specified task. In contrast, task-agnostic grasp approaches give the same grasp for different tasks. Figure 3 shows how a hammer might be grasped with a task-agnostic and a task-oriented approach: While the task-agnostic grasp is the same among all tasks, the task-oriented grasps differ based on the required task. The task-oriented approach aims to understand the object context and grasps differently based on the instructed task. One common approach is to learn how humans use a specific object. Grasping is not the end of the sequence, task execution comes afterwards and therefore the robot needs to understand the object context and find a grasp that is sufficient for the given task. Task-oriented approaches receive more attention recently [22, 24, 27]. We will focus on the task-oriented approaches as they spark potential for the future and are closer to the vision of robots executing complex task chains unsupervised. In addition, multiple grippers have been used in the past. Earlier research used a static two-finger parallel jaw for gripping, while recent research often uses a dynamic three- or five-finger gripper [20, 23, 25].

This paper focuses on 6-DoF task-oriented grasping (TOG). By utilizing 6-DoF, approaches can ensure a broader adaptability to different environments and tasks, thus improving the practical applications of robot grasping in real-world applications. TOG comes closer to the vision of robots performing complex task chains in an unsupervised manner. The paper is organized as follows: Section 2 outlines related research, section 3 details the challenges of TOG, and section 4 explains the most common metrics for evaluating grasping. In addition, section 5 details two con-

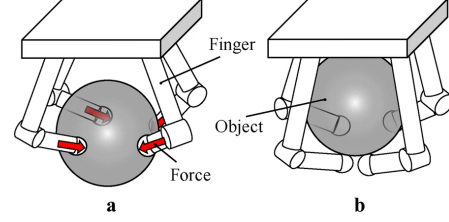


Figure 4. Grasp movement restraint by force closure (a) and by form closure (b). Image taken from [14].

crete approaches and how they address certain challenges. Finally, section 6 provides an outlook on the future directions of TOG.

2. Related Work

This section aims to overview past work in the field of TOG. Therefore, evaluation criteria for grasp candidates are described and the general methods are explained.

Grasp Synthesis evaluates grasp candidates to find the most stable grasp. Two common evaluation criteria are force closure and form closure. As shown in the figure 4: Force closure is achieved by restricting any movement of the object due to the force applied to the contact points. On the other hand, form closure is achieved by the geometric arrangement of contact points that completely constrain the object's movement, regardless of external forces. Grasps with multiple contacts between the object and the fingers and palm are considered power grasps because they provide a high load carrying capacity [9, 20]. In addition, task-oriented grasp synthesis evaluates how suitable a grasp candidate is for the requested task. There is no single solution for measuring the success of a task-oriented grasp. Therefore, we will discuss example evaluations in a later section [20, 23, 24].

In earlier research, analytic approaches were preferred over data-driven approaches. Analytical approaches assume full knowledge of object shape, geometry, and material properties. These approaches analyze the shape of an object to find an appropriate grasp pose. The complete information about shape, geometry and material properties is often not available. Therefore, data-driven approaches have emerged to address the missing information and higher complexity of environments. They often incorporate machine learning and use the knowledge of previous successful grips. In addition, data-driven approaches can be applied to unknown objects and grasping simulator such as "GraspIt!" support the rise of data-driven approaches. For these reasons, recent research uses data-driven approaches more [9, 19, 25]. Recent data-driven approaches learn TOG from RGB or RGB-D observations without requiring additional object properties [19, 26].

In recent research, approaches are often data-driven and

include a Deep Learning (DL) network. DL is utilized to train large neural networks and has got more attention in the last years. Therefore, in the next paragraph, four main methods within the DL domain will be described in detail to lay the groundwork for the concrete approach described in section 5. Prior to 2020, the predominant approach to grasp synthesis was sampling-based methods. These methods work by individually extracting samples from a discrete or continuous high-dimensional space, evaluating them based on a learned quality metric, and optionally refining the samples for optimization. A commonly used refinement involves sampling from a latent space, which can improve the quality of the grasp, but often comes with additional computational overhead [18, 20]. Direct regression methods use a single neural network to process the entire sample space in one pass, predicting single or multiple grasps and their properties from visual input. These end-to-end approaches aim to reduce computational costs compared to sampling methods by processing data globally. Some direct regression methods achieve real-time performance. While direct regression enables fast inference for tasks such as closed-loop and dynamic object grasping, scaling to high-dimensional outputs such as 6-DOF environments is challenging. To address this, researchers use multi-stage approaches or reduce output complexity by conditioning predictions on input constraints [20, 25]. Reinforcement learning (RL) grasping methods aim to maximise a cumulative reward function, often over multi-step tasks. To do this, an agent uses trial and error to maximise the reward function. RL approaches typically generate a single grasp per scene, making it difficult to enforce additional constraints during execution [12, 20]. Finally, the exemplar methods use a learning of similarity metrics between grasps to determine high quality grasps from an existing database. Therefore, previous successful grasps are used for the grasp context. Exemplar methods are limited to the knowledge of the database and can therefore hardly grasp objects that are significantly different from the seen object. However, some exemplar methods partially overcome this knowledge by reducing the target object to the most similar known object [12, 16, 20].

3. Challenges

This section aims to outline the challenges in the area of TOG, ranging from task context awareness, limited available data and visual occlusion of objects, to unclear generalization quality.

The human hand has 24 DoFs and each DoF is not independent. This makes grasping a complex problem. To grasp like a human, the robot must understand the purpose of the object under different circumstances. For example, if a hammer is not available, a human might pick up a book and use it as a hammer [3, 7, 9, 20]. In addition, it is unclear

how to measure the success of a task-oriented grasp. Different grasping criteria need to be designed for different task-oriented grasps [9]. Another challenge is grasping objects with limited visual information. For example, partial occlusion of an object in a cluttered scene can cause algorithms to struggle to find an appropriate grasp [6, 23, 25, 27]. With the growth of supervised deep learning, data labeling becomes a challenging problem. For TOG, the availability of labeled data is limited, and due to the complexity of TOG with different tasks, it is difficult to annotate the data [11, 15, 26]. Furthermore, it is difficult for the algorithms to generalize to novel objects and novel tasks. Each algorithm is trained on a fixed set of tasks and objects, which makes translating the information to previously unseen objects and tasks challenging [13, 23]. In addition, most approaches are trained in simulation due to reduced computational cost and faster training. As a result, performance in reality drops significantly [4, 17].

4. Evaluation

This section provides an overview of some of the most commonly used datasets and evaluation metrics for benchmarking TOG systems.

4.1. Datasets

Table 1 shows some of the most widely used datasets in TOG. Note, that TOG-Net provides the highest number of objects and grasps, but the variety of objects and tasks is low. An algorithm solely trained on TOG-Net might have limited generalization ability due to that. A similar limitation is given when using ContactDB as the number of tasks is two. Therefore, TaskGrasp and LaViA-TaskGrasp provide the most variety in terms of object categories and tasks. LaViA-TaskGrasp is the TaskGrasp dataset enriched with additional language and visual data. TaskGrasp used crowdsourcing and majority voting of highly accurate annotators to label the task-object combinations.

	ContactDB [5]	TOG-Net [10]	TaskGrasp [19]	LaViA-TaskGrasp [23]
Semantic Knowledge	×	×	✓	✓
Multimodal (Language + Vision)	×	×	×	✓
Object Categories	40	2	75	75
Objects	50	18K (synthetic)	191	191
Tasks	2	2	56	56
Grasps	3750	1.5M	250K	250K

Table 1. Comparison of datasets for TOG.

4.2. Metrics

Benchmarking TOG systems require a set of metrics that assess both the efficiency and effectiveness of a grasping strategy. The following metrics are commonly used in the literature to benchmark grasping performance. **Grasp Task Success Rate** measures the percentage of successful grasps over the total number of grasp attempts. A grasp is considered successful if it holds the object reliably and enables

the intended task to be performed. **Computation Time** is the time required to compute candidate grasps. This metric evaluates the efficiency of the grasp prediction algorithm. **Coverage** measures how well the predicted grasps match the ground truth grasps. It is defined as the percentage of sampled ground truth grasps that are close to one of the generated grasp candidates. **Precision** measures the accuracy of a model in predicting valid grips. It is defined as the percentage of true positive grasp predictions over the total number of selected grasp candidates. **Accuracy** evaluates the quality of the predictions by calculating the percentage of successful predictions divided by the total number of predictions. The naming of the metrics may differ between papers. However, these are among the most common metrics for benchmarking TOG systems [19, 20].

5. Approaches

5.1. CaTGrasp [26]

This section aims to provide an overview of an exemplar data-driven method in the domain of industrial assembly. It outlines how Wen et al. approached the challenges mentioned in section 3 and briefly discusses future improvements to their work.

CatGrasp uses a 2-finger parallel gripper in 6-DoF for industrial assembly tasks. The algorithm grasps an object from a cluttered scene and then performs a placement task. Figure 5 illustrates the approach. To overcome the challenge of computationally expensive data collection and labeling, the authors use synthetic unlabeled data from point clouds of singulated objects. Wen et al. simulate different grips on each object and evaluate success by lifting and placing the object at the target location. Learning from synthetic simulation and learning from human demonstrations are two common solutions for avoiding manual labelling [23]. Their simulation process yields a representation of semantic information, including shape, pose, and gripper-object contact points for each category. During execution, they extract a target object from the visual point cloud input and transfer the corresponding category-level semantic information to the target object. Therefore, the approach can be classified into the exemplar methods as it transfers knowledge from a database based on object similarity. Then, the approach samples multiple grasp candidates and applies the two metrics of grasp stability and successful object placement to obtain the best grasp. In addition to traditional grasp stability measures, success is also defined by how well the object is placed after the grasp. This is an example of how domain knowledge is used to define the success of a grasp. In other domains, such as house-holding, there are objects and tasks for which successful placement is not an appropriate measurement. Their paper shows that this approach performs nearly as well in real-

world scenarios as it does in simulations. However, since they only use three categories, future work could expand the number of categories or even use a solution that generalizes well to previously unseen objects.

5.2. FoundationGrasp [23]

In the following section, we provide an overview of an approach that generalizes more effectively to unseen objects and tasks. Key components and concepts of the algorithm are explained, along with a brief discussion of its limitations.

Tang et. al. use an LLM integration to improve the generalization capability of TOG. They show that their approach improves generalization to novel unseen objects and novel unseen tasks. Objects that differ semantically such as "spoon" and "saucepan" can share a similar geometric. Therefore, the awareness of similar geometric information enables the approach to generalize to semantically novel objects. They use the LaViA-TaskGrasp dataset mentioned in section 4.1, which contains a variety of kitchen objects and kitchen tasks. To evaluate a given grasp candidate, the approach consists of the following three steps:

1. **Knowledge generation:** The learning approach is based on a semantic and geometric information branch. In the first stage, the information required by the encoder is generated. Namely, the input consists of an image of the object, a language instruction, and a grasping candidate given together with the object's point cloud. The language instruction is a task-object combination such as "pick up the cup to pour". In the next step an LLM, in this case OpenAI GPT-4 [1], is prompted to provide a geometric description of the object and the task. For example the object "Mug" leads to the description "A mug is composed of a cylindrical body, a handle, a circular base, and often features an outer glaze or design". To overcome the issue of occlusion and limited visual information, the visual information is enriched with crowdsource images from the internet that provides different views on the object. Geometric descriptions and visual information are given to the geometric branch. On the semantic information side, the LLM is prompted to provide a description of the task. The language instruction and the semantic description are the input to the semantic branch.
2. **Multi-modal feature representation:** The generated information is then encoded by a semantic and a geometric knowledge encoder. A pre-trained BERT [8] is used to encode the semantic information. Since the geometric encoder combines visual and textual information, preprocessing is required. A visual language model, namely OpenAI CLIP [21], is used to extract

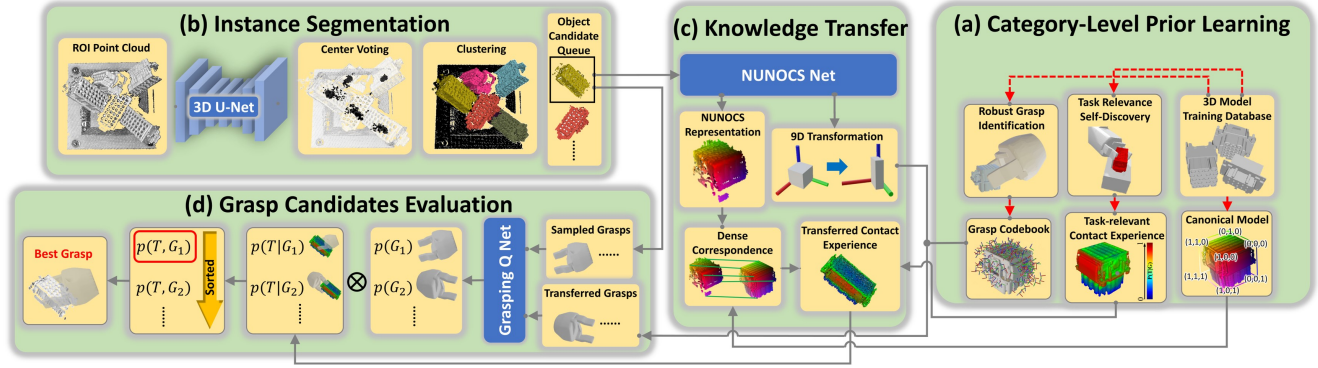


Figure 5. Overview of CaTGrasp. (a) Category-Level information of objects and task-relevant contact points is learned prior to execution. (b) Given a point cloud the objects are clustered and a target object is chosen. (c) The Category-Level prior knowledge is transferred to the target object. (d) Grasp candidates are sampled based on the transferred knowledge and based on the given point cloud. Grasp candidates are evaluated and the best grasp is selected. Image taken from [26].

multi-modal features from the input image. The geometric information is then encoded.

3. **Task-oriented grasp evaluation:** The last step of the approach is the transformer-based task-oriented grasp evaluation. It takes the encoded geometric and semantic information and processes them separately. It uses multi-head cross-attention and self-attention mechanisms to transform the novel given elements into a known one. Finally, the outputs of a mean pooling on the geometric and semantic parts are concatenated and translated through a multilayer perceptron and a sigmoid function into a number from 0 to 1. This value can be understood as the quality of the grasp candidate.

Finally, Tang et al. sample several grasps with a task-agnostic sampler and evaluate each one with the task-oriented grasp evaluator described above. The language input leads to the extraction of a target object and, using the described framework, to a best grasp. Due to the LLM description of task and object, the approach performs significantly better in generalization. When the language instructions mention to "Sweep the table", the semantic description contains information about sweeping and is able to find similar verbs. A limitation of their approach is that data collection requires labor-intensive manual annotation. The authors report that FoundationGrasp prefers grasp poses near the end of the handle. These grasps are less robust to perturbations. Also, as they focus on the two-finger gripper, the approach cannot be directly applied on other grippers such as the five-finger hand.

6. Outlook

In this work, we briefly overviewed TOG in 6-DoF. From the pipeline, over the advantages of TOG to the challenges of the field. Additionally, we described two approaches

and how they overcome certain challenges. While we overviewed certain benchmarking metrics, a standardized benchmarking process would help to compare approaches more effectively. The code for both approaches is publicly available. This enables other researchers to build up on the code and drives progress in research. However, as a lot of papers do not make their code publicly available, it should become more common to publish the code along with the paper. Also, research in this area has been mostly around 2-finger grippers, the slowly starting trend of 3- and 5-finger grippers should continue. The more finger hand, especially the 5-finger hand, come closer to human grasping and enables a wider range of grasps. Furthermore, researchers need to consider the entire robotic pipeline: TOG should be researched more and a combination with language input, gesture communication or other human-robot interaction topics might lead to a closer alignment with applications of robots in day-to-day life.

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